

Solution Sheet 2— The Stochastic Integral

2.1: We want to write the integral as the limit of integrals of simple processes, so let the partition Π^n have points $t_j^n = \frac{jt}{n}$ and set $\phi_s^n = \sum_{j=0}^{n-1} t_j^n \mathbf{1}_{[t_j^n, t_{j+1}^n)}(s)$. We can see that this sequence of simple processes converges in the appropriate sense to t , and so

$$\begin{aligned} \int_0^t s dW_s &= \lim_{n \rightarrow \infty} \int_0^t \phi_s^n dW_s \\ &= \lim_{n \rightarrow \infty} \sum_{j=0}^{n-1} t_j^n (W_{t_{j+1}^n} - W_{t_j^n}). \end{aligned}$$

Now note that we can write $t_{j+1}^n W_{t_{j+1}^n} - t_j^n W_{t_j^n} = t_j^n (W_{t_{j+1}^n} - W_{t_j^n}) + W_{t_{j+1}^n} (t_{j+1}^n - t_j^n)$, and so

$$\begin{aligned} \int_0^t s dW_s &= \lim_{n \rightarrow \infty} \sum_{j=0}^{n-1} [t_{j+1}^n W_{t_{j+1}^n} - t_j^n W_{t_j^n} - W_{t_{j+1}^n} (t_{j+1}^n - t_j^n)] \\ &= tW_t - \int_0^t W_s ds. \end{aligned}$$

2.2: (a) We can rewrite $\int_0^T W_t dt$ as a limit of sums along the same partition, and then we have

$$\begin{aligned} \lim \sum_i W_{t_i} (W_{t_{i+1}} - W_{t_i})^2 - \int_0^T W_t dt \\ &= \lim \sum_i \left[W_{t_i} (W_{t_{i+1}} - W_{t_i})^2 - W_{t_i} (t_{i+1} - t_i) \right] \\ &= \lim \sum_i \left[W_{t_i} ((W_{t_{i+1}} - W_{t_i})^2 - t_{i+1} - t_i) \right], \end{aligned}$$

so

$$\begin{aligned} &\left(\sum_i W_{t_i} (W_{t_{i+1}} - W_{t_i})^2 - \int_0^T W_t dt \right)^2 \\ &= \sum_i W_{t_i}^2 (W_{t_{i+1}} - W_{t_i})^4 + \sum_i W_{t_i}^2 (t_{i+1} - t_i)^2 - 2 \sum_i W_{t_i}^2 (W_{t_{i+1}} - W_{t_i})^2 (t_{i+1} - t_i) \\ &\quad + \sum_{i < j} W_{t_i} (W_{t_{i+1}} - W_{t_i})^2 W_{t_j} (t_{j+1} - t_j). \end{aligned}$$

Now,

$$\begin{aligned} \mathbb{E} [W_{t_i}^2 (W_{t_{i+1}} - W_{t_i})^4] &= \mathbb{E} [W_{t_i}^2] \mathbb{E} [(W_{t_{i+1}} - W_{t_i})^4] = 3t_i (t_{i+1} - t_i)^2, \\ \mathbb{E} [W_{t_i}^2 (t_{i+1} - t_i)^2] &= t_i (t_{i+1} - t_i)^2, \end{aligned}$$

and

$$\begin{aligned} \mathbb{E} [W_{t_i}^2 (W_{t_{i+1}} - W_{t_i})^2 (t_{i+1} - t_i)] &= \mathbb{E} [W_{t_i}^2] \mathbb{E} [(W_{t_{i+1}} - W_{t_i})^2] (t_{i+1} - t_i) \\ &= t_i (t_{i+1} - t_i)^2. \end{aligned}$$

For the cross terms we have (for $i < j$)

$$\begin{aligned}
& \mathbb{E} \left[W_{t_i} \left((W_{t_{i+1}} - W_{t_i})^2 - t_{i+1} - t_i \right) W_{t_j} \left((W_{t_{j+1}} - W_{t_j})^2 - t_{j+1} - t_j \right) \right] \\
&= \mathbb{E} \left[\mathbb{E} \left[W_{t_i} \left((W_{t_{i+1}} - W_{t_i})^2 - t_{i+1} - t_i \right) W_{t_j} \left((W_{t_{j+1}} - W_{t_j})^2 - t_{j+1} - t_j \right) \middle| \mathcal{F}_{t_j} \right] \right] \\
&= \mathbb{E} \left[W_{t_i} \left((W_{t_{i+1}} - W_{t_i})^2 - t_{i+1} - t_i \right) W_{t_j} \mathbb{E} \left[(W_{t_{j+1}} - W_{t_j})^2 - t_{j+1} - t_j \middle| \mathcal{F}_{t_j} \right] \right] \\
&= 0.
\end{aligned}$$

Combining these, we get that

$$\begin{aligned}
\mathbb{E} \left[\left(\sum_i W_{t_i} (W_{t_{i+1}} - W_{t_i})^2 - W_{t_i} (t_{i+1} - t_i) \right)^2 \right] &= 2 \sum_i t_i (t_{i+1} - t_i)^2 \\
&\leq \|\Pi\| T \sum_i (t_{i+1} - t_i) \\
&= \|\Pi\| T^2 \\
&\rightarrow 0, \quad \text{as } \|\Pi\| \rightarrow 0.
\end{aligned}$$

(b) We take a similar approach to above and find that

$$\begin{aligned}
& \mathbb{E} \left[\left(\sum_i (W_{t_{i+1}} - W_{t_i})^3 \right)^2 \right] \\
&= \sum_i \mathbb{E} \left[(W_{t_{i+1}} - W_{t_i})^6 \right] + 2 \sum_{i < j} \mathbb{E} \left[(W_{t_{i+1}} - W_{t_i})^3 (W_{t_{j+1}} - W_{t_j})^3 \right] \\
&= 15 \sum_i (t_{i+1} - t_i)^3 + \sum_{i < j} \mathbb{E} \left[\mathbb{E} \left[(W_{t_{i+1}} - W_{t_i})^3 (W_{t_{j+1}} - W_{t_j})^3 \middle| \mathcal{F}_{t_j} \right] \right] \\
&= 15 \sum_i (t_{i+1} - t_i)^3 + \sum_{i < j} \mathbb{E} \left[(W_{t_{i+1}} - W_{t_i})^3 \mathbb{E} \left[(W_{t_{j+1}} - W_{t_j})^3 \middle| \mathcal{F}_{t_j} \right] \right] \\
&= 15 \sum_i (t_{i+1} - t_i)^3 \\
&\leq 15 \|\Pi\|^2 T \\
&\rightarrow 0, \quad \text{as } \|\Pi\| \rightarrow 0.
\end{aligned}$$

(c) To use the previous two parts we note

$$(W_{t_{i+1}} - W_{t_i})^3 = W_{t_{i+1}}^3 - W_{t_i}^3 - 3W_{t_i}^2 (W_{t_{i+1}} - W_{t_i}) - 3W_{t_i} (W_{t_{i+1}} - W_{t_i})^2,$$

so that

$$\begin{aligned}
3 \int_0^t W_s^2 dW_s &= 3 \lim \sum_i W_{t_i}^2 (W_{t_{i+1}} - W_{t_i}) \\
&= \lim \sum_i \left[(W_{t_{i+1}}^3 - W_{t_i}^3) - (W_{t_{i+1}} - W_{t_i})^3 - 3W_{t_i} (W_{t_{i+1}} - W_{t_i})^2 \right] \\
&= W_t^3 + 0 - 3 \int_0^t W_s ds \quad (\text{by the previous parts}) \\
&= W_t^3 - 3 \int_0^t W_s ds.
\end{aligned}$$

- 2.3:** *i)* Take the simple process $\phi_r = \sum_{j=0}^{N-1} e_j \mathbf{1}_{[t_j, t_{j+1})}(r)$, where $t_0 \leq s$ and $t_N \geq t$. Then $s \in [t_k, t_{k+1})$, $t \in [t_l, t_{l+1})$ for some k, l , and so $\phi_r \mathbf{1}_{[s, t)}(r) = \sum_{j=0}^{M-1} e_j \mathbf{1}_{[\tilde{t}_j, \tilde{t}_{j+1})}(r)$, where $\tilde{t}_0 = s, \tilde{t}_1 = t_{k+1}, \dots, \tilde{t}_{M-1} = t_l, \tilde{t}_M = t$. If $u \in [\tilde{t}_i, \tilde{t}_{i+1})$ then

$$\begin{aligned}
 \int_s^t \phi_r dW_r &= \int_0^\infty \phi_r \mathbf{1}_{[s, t)}(r) dW_r \\
 &= \sum_{j=0}^{M-1} e_j (W_{\tilde{t}_{j+1}} - W_{\tilde{t}_j}) \\
 &= \sum_{j=0}^{i-1} e_j (W_{\tilde{t}_{j+1}} - W_{\tilde{t}_j}) + e_i (W_u - W_{\tilde{t}_i}) + e_i (W_{\tilde{t}_{i+1}} - W_u) \\
 &\quad + \sum_{j=i+1}^{M-1} e_j (W_{\tilde{t}_{j+1}} - W_{\tilde{t}_j}) \\
 &= \sum_{j=0}^i e_j (W_{\tilde{t}_{j+1}} - W_{\tilde{t}_j}) + \sum_{j=i+1}^M e_j (W_{\tilde{t}_{j+1}} - W_{\tilde{t}_j}) \\
 &= \int_0^\infty \phi_r \mathbf{1}_{[s, u)}(r) dW_r + \int_0^\infty \phi_r \mathbf{1}_{[u, t)}(r) dW_r.
 \end{aligned}$$

- ii)* If $\psi_r \mathbf{1}_{[s, t)}(r) = \sum_{j=0}^{N-1} f_j \mathbf{1}_{[\tilde{s}_j, \tilde{s}_{j+1})}(r)$, then we can combine the partitions of the two simple processes to get a new partition of times $\hat{t}_j$, and then

$$\begin{aligned}
 \int_s^t (\alpha \phi_r + \beta \psi_r) dW_r &= \int_0^\infty (\phi_r \mathbf{1}_{[s, t)}(r) + \psi_r \mathbf{1}_{[s, t)}(r)) dW_r \\
 &= \sum_{j=0}^{\hat{M}-1} (\alpha e_j + \beta f_j) (W_{\hat{t}_{j+1}} - W_{\hat{t}_j}) \\
 &= \alpha \sum_{j=0}^{M-1} e_j (W_{\tilde{t}_{j+1}} - W_{\tilde{t}_j}) + \beta \sum_{j=0}^{N-1} f_j (W_{\tilde{s}_{j+1}} - W_{\tilde{s}_j}) \\
 &= \alpha \int_s^t \phi_r dW_r + \beta \int_s^t \psi_r dW_r.
 \end{aligned}$$

- iii)* Also,

$$\begin{aligned}
 \mathbb{E} \left[\int_s^t \phi_r dW_r \middle| \mathcal{F}_s \right] &= \mathbb{E} \left[\sum_{j=0}^{M-1} e_j (W_{\tilde{t}_{j+1}} - W_{\tilde{t}_j}) \middle| \mathcal{F}_s \right] \\
 &= \sum_{j=0}^{M-1} \mathbb{E} \left[\mathbb{E} \left[e_j (W_{\tilde{t}_{j+1}} - W_{\tilde{t}_j}) \middle| \mathcal{F}_{\tilde{t}_j} \right] \middle| \mathcal{F}_s \right] \\
 &= \sum_{j=0}^{M-1} \mathbb{E} \left[e_j \mathbb{E} \left[W_{\tilde{t}_{j+1}} - W_{\tilde{t}_j} \middle| \mathcal{F}_{\tilde{t}_j} \right] \middle| \mathcal{F}_s \right] \quad (e_j \text{ is known at time } \tilde{t}_j) \\
 &= 0.
 \end{aligned}$$

$$\text{Also } \mathbb{E} \left[\int_s^t \phi_r dW_r \right] = \mathbb{E} \left[\mathbb{E} \left[\int_s^t \phi_r dW_r \middle| \mathcal{F}_s \right] \right] = 0.$$

iv) If $\phi_r \mathbf{1}[0, t)(r) = \sum_{j=0}^{N-1} e_j \mathbf{1}_{[t_j, t_{j+1})}(r)$, then $\int_0^t \phi_r dW_r = \sum_{j=0}^{N-1} e_j (W_{t_{j+1}} - W_{t_j})$, but each e_j is $\mathcal{F}_{t_j}$ -measurable, and each $W_{t_{j+1}} - W_{t_j}$ is $\mathcal{F}_{t_{j+1}}$ -measurable. Then, since $\mathcal{F}_{t_j} \subseteq \mathcal{F}_{t_{j+1}} \subseteq \mathcal{F}_t$, we have that for each t , $\int_0^t \phi_r dW_r$ is $\mathcal{F}_t$ -measurable, and therefore the process is adapted.

For continuity note that

$$\begin{aligned} \int_0^{t+\varepsilon} \phi_r dW_r &= \sum_{j=0}^N e_j (W_{t_{j+1}} - W_{t_j}) \\ &= \sum_{j=0}^{N-1} e_j (W_{t_{j+1}} - W_{t_j}) + e_N (W_{t_{N+1}} - W_{t_N}) \\ &= \int_0^t \phi_r dW_r + e_N (W_{t+\varepsilon} - W_t) \\ &\rightarrow \int_0^t \phi_r dW_r \quad \text{as } \varepsilon \rightarrow 0. \end{aligned}$$

Finally, we observe that these properties tend to be observed in the limit, so:

- **Additivity:** If φ^N is a sequence of simple functions approximating $\varphi \in \mathcal{V}$, then *a fortiori* $\varphi^N \mathbf{1}_{\{t\}}[s, u)$ approximate $\varphi \mathbf{1}_{\{t\}}[s, u)$ and hence $\int_s^u \varphi_r^N dW_r \rightarrow \int_s^u \varphi_r dW_r$ in $\mathcal{L}^2(\mathbb{P})$. Since similar convergence holds for all the relevant terms, we must get the required identity in the limit.
- **Linearity:** Similar to above, $\int_s^t \varphi_r^N dW_r \rightarrow \int_s^t \varphi_r dW_r$, etc., so we get equality of the limiting sum.
- **Expectation:** This follows from the additional fact that convergence in $\mathcal{L}^2(\mathbb{P})$ implies convergence in $\mathcal{L}^1(\mathbb{P})$, and $\mathbb{E}[|X_n - X|] \rightarrow 0$ implies $\mathbb{E}[|X_n|] \rightarrow \mathbb{E}[|X|]$. (By Jensen's inequality $|\mathbb{E}[|X_n|] - \mathbb{E}[|X|]| \leq \mathbb{E}[||X_n| - |X||]$, and by the reverse triangle inequality, $\mathbb{E}[||X_n| - |X||] \leq \mathbb{E}[|X_n - X|]$.)
- **Adaptedness:** The limit is adapted, since the limit of adapted random variables is itself adapted.

2.4: It is easy to verify that

$$\begin{aligned} h_0(x) &= 1 \\ h_1(x) &= x \\ h_2(x) &= x^2 - 1 \\ h_3(x) &= x^3 - 3x. \end{aligned}$$

For $n = 1$ we have

$$\int_0^t dW_u = W_t = \sqrt{t} \frac{W_t}{\sqrt{t}} = t^{\frac{1}{2}} h_1\left(\frac{W_t}{\sqrt{t}}\right),$$

as required. Then,

$$\begin{aligned}
 \int_0^t \int_0^{u_2} dW_{u_1} dW_{u_2} &= \int_0^t W_{u_2} dW_{u_2} \\
 &= \frac{1}{2} W_t^2 - \frac{1}{2} t \quad (\text{from the example used in lecture notes}) \\
 &= \frac{t}{2!} \left(\frac{W_t^2}{t} - 1 \right) \\
 &= \frac{t}{2!} h_2 \left(\frac{W_t}{\sqrt{t}} \right).
 \end{aligned}$$

Finally,

$$\begin{aligned}
 \int_0^t \int_0^{u_3} \int_0^{u_2} dW_{u_1} dW_{u_2} dW_{u_3} &= \int_0^t \left(\frac{1}{2} W_{u_3}^2 - \frac{1}{2} u_3 \right) dW_{u_3} \\
 &= \frac{1}{6} W_t^3 - \frac{1}{2} \int_0^t W_s dW_s - \frac{1}{2} \int_0^t s dW_s \quad (\text{by Q2.2}) \\
 &= \frac{1}{6} W_t^3 + \frac{1}{2} \left(\int_0^t s dW_s - t W_t \right) - \frac{1}{2} \int_0^t s dW_s \quad (\text{by Q2.1}) \\
 &= \frac{1}{6} W_t^3 - \frac{1}{2} t W_t \\
 &= \frac{t^{3/2}}{3!} h_3 \left(\frac{W_t}{\sqrt{t}} \right).
 \end{aligned}$$

2.5: Notice that

$$\begin{aligned}
 2 \int_0^t \phi_u dW_u \int_0^t \psi_u dW_u &= \left(\int_0^t \phi_u dW_u - \int_0^t \psi_u dW_u \right)^2 - \left(\int_0^t \phi_u dW_u \right)^2 \\
 &\quad - \left(\int_0^t \psi_u dW_u \right)^2,
 \end{aligned}$$

so that, by the Itô Isometry,

$$\begin{aligned}
 \mathbb{E} \left[2 \int_0^t \phi_u dW_u \int_0^t \psi_u dW_u \right] &= \mathbb{E} \left[\int_0^t (\phi_u - \psi_u)^2 du - \int_0^t \phi_u^2 du - \int_0^t \psi_u^2 du \right] \\
 &= \mathbb{E} \left[\int_0^t [(\phi_u - \psi_u)^2 - \phi_u^2 - \psi_u^2] du \right] \\
 &= \mathbb{E} \left[\int_0^t 2\phi_u \psi_u du \right].
 \end{aligned}$$

2.6: By definition,

$$\begin{aligned}
 \int_0^t W_u \circ dW_u &= \lim \sum_i \frac{1}{2} (W_{t_{i+1}} + W_{t_i}) (W_{t_{i+1}} - W_{t_i}) \\
 &= \lim \sum_i \frac{1}{2} (W_{t_{i+1}}^2 - W_{t_i}^2) \\
 &= \frac{1}{2} W_t^2.
 \end{aligned}$$

Similarly,

$$\begin{aligned}
\int_0^t W_u^2 \circ dW_u &= \lim \sum_i \frac{1}{2} (W_{t_{i+1}}^2 + W_{t_i}^2) (W_{t_{i+1}} - W_{t_i}) \\
&= \lim \sum_i \left[\frac{1}{2} (W_{t_{i+1}} - W_{t_i})^3 + W_{t_i} (W_{t_{i+1}} - W_{t_i}) + W_{t_i} (W_{t_{i+1}} - W_{t_i}) \right] \\
&= 0 + \int_0^t W_u du + \int_0^t W_u^2 dW_u \\
&= \frac{1}{3} W_t^3,
\end{aligned}$$

by Q2.2.

2.7: We can write

$$\int_0^T \phi_u dW_u - \lim_{\Pi} \sum_i \phi_{t_i}^* (W_{t_{i+1}} - W_{t_i}) = \lim_{\Pi} \sum_i (\phi_{t_i} - \phi_{t_i}^*) (W_{t_{i+1}} - W_{t_i}),$$

but by the Cauchy-Schwartz inequality,

$$\begin{aligned}
\mathbb{E} \left[\left| \sum_i (\phi_{t_i} - \phi_{t_i}^*) (W_{t_{i+1}} - W_{t_i}) \right|^2 \right] &\leq \mathbb{E} \left[\sum_i (\phi_{t_i} - \phi_{t_i}^*)^2 \right] \mathbb{E} \left[\sum_i (W_{t_{i+1}} - W_{t_i})^2 \right] \\
&= \sum_i \mathbb{E} [(\phi_{t_i} - \phi_{t_i}^*)^2] \sum_j \mathbb{E} [(W_{t_{j+1}} - W_{t_j})^2] \\
&\leq \sum_i (t_i^* - t_i)^{1+\varepsilon} \sum_j (t_{j+1} - t_j) \\
&\leq T \sum_i (t_{j+1} - t_j)^{1+\varepsilon} \\
&\leq T^2 \|\Pi\|^\varepsilon \\
&\rightarrow 0 \quad \text{as } \|\Pi\| \rightarrow 0.
\end{aligned}$$

We have therefore shown convergence in $\mathcal{L}^1(\mathbb{P})$, and convergence in probability follows.

2.8: (a) We have already discussed adaptedness and continuity, so it remains to show that we have integrability and the martingale property. For integrability note that by Jensen's Inequality, the Itô Isometry, and the definition of $\mathcal{V}$,

$$\left(\mathbb{E} \left[\left| \int_0^t \phi_s dW_s \right| \right] \right)^2 \leq \mathbb{E} \left[\left(\int_0^t \phi_s dW_s \right)^2 \right] = \mathbb{E} \left[\int_0^t \phi_s^2 ds \right] < \infty.$$

For the martingale property,

$$\begin{aligned}
 \mathbb{E} \left[\int_0^t \phi_u dW_u \middle| \mathcal{F}_s \right] &= \mathbb{E} \left[\int_0^s \phi_u dW_u + \int_s^t \phi_u dW_u \middle| \mathcal{F}_s \right] \\
 &= \int_0^s \phi_u dW_u + \mathbb{E} \left[\sum_{s \leq t_j \leq t_{j+1} \leq t} e_j (W_{t_{j+1}} - W_{t_j}) \middle| \mathcal{F}_s \right] \\
 &= \int_0^s \phi_u dW_u + \sum_{s \leq t_j \leq t_{j+1} \leq t} \mathbb{E} \left[\mathbb{E} \left[e_j (W_{t_{j+1}} - W_{t_j}) \middle| \mathcal{F}_{t_j} \right] \middle| \mathcal{F}_s \right] \\
 &= \int_0^s \phi_u dW_u.
 \end{aligned}$$

- (b) We have shown that for each n , I^n is a martingale, and therefore $I^n - I^m$ is also a martingale for any n, m , so we can use Doob's Martingale Inequality on this. Then, by the Ito Isometry again,

$$\begin{aligned}
 \mathbb{P} \left(\sup_{0 \leq t \leq T} |I_t^n - I_t^m| \geq \lambda \right) &\leq \frac{1}{\lambda^2} \mathbb{E} [|I_T^n - I_T^m|^2] \\
 &= \frac{1}{\lambda^2} \mathbb{E} \left[\int_0^T (I_t^n - I_t^m)^2 dt \right] \\
 &\rightarrow 0 \quad \text{as } m, n \rightarrow \infty.
 \end{aligned}$$

Therefore we can choose a subsequence n_k such that

$$\mathbb{P} \left(\sup_{0 \leq t \leq T} |I_t^{n_{k+1}} - I_t^{n_k}| > 2^{-k} \right) < 2^{-k}.$$

- (c) The Borel-Cantelli Lemma then tells us that

$$\mathbb{P} \left(\sup_{0 \leq t \leq T} |I_t^{n_{k+1}} - I_t^{n_k}| > 2^{-k} \text{ for infinitely many } k \right) = 0,$$

so for almost every ω there is a $k_1(\omega)$ such that

$$\sup_{0 \leq t \leq T} |I_t^{n_{k+1}}(\omega) - I_t^{n_k}(\omega)| \geq 2^{-k} \quad \text{for } k \geq k_1(\omega).$$

Therefore $I_t^{n_k}(\omega)$ is uniformly convergent for $0 \leq t \leq T$ and almost every ω , so the limit is t -continuous for $0 \leq t \leq T$ almost surely. But $I_t^{n_k} \rightarrow I_t$ in $\mathcal{L}^2(\mathbb{P})$ for all t , and we are done.